

cryptography are based on the properties and the characteristics of these photons. In digital signal processing, a bit represents the state of 0 or 1 at a particular instant; quantum annealing defines a bit in three states of 0, 1 and in superposition state (simultaneous occurrence or orientation of both 0 and 1). The states of the qubit are represented in terms of probabilistic amplitudes, though different representations of the qubit are adopted by various research firms, the most common and standard being the ‘ket representation’, 0 as $|0\rangle$ and 1 as $|1\rangle$. In general, the state of a qubit is defined on the complex plane by the equation: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$ and are called the amplitudes. The probability of the occurrence of $|0\rangle$ is $|\alpha|^2$ and of $|1\rangle$ is $|\beta|^2$. The set of all possible orientations of $|\psi\rangle$ is plotted on Hilbert’s space (a complex unit circle). Often, a different geometrical representation (Bloch sphere) is done for qubits that have pure states or a two level quantum state. Discussing Hilbert’s space and Bloch sphere is beyond the scope of this article, so interested readers can contact the author via email. Figure 1 shows a normal bit plot and Figure 2 shows a qubit plot.

Measurement of a qubit: Entanglement is a property specifically related to quantum mechanics, which increases the degree of co-relation among qubits. According to this property, if the quantum state of one qubit is known, then the quantum state of the other qubits in the same reference frame can be found. For example, in a hypothetical reference frame, if there are two qubits and if we know that one qubit is spinning clockwise, then due to analogous behaviour, the other qubit must spin in the anticlockwise direction. The latter qubit is called the mate of the former. Due to entanglement, qubits achieve inherent parallelism.

As discussed before, qubits are the abstraction of photons with their control devices, and due to their potential to incorporate multiple states simultaneously, they are capable of processing millions of computations in a single instant. A single qubit, due to its superposition, is capable of carrying the two states, 0 and 1, in a single instant; hence, what two traditional bits can do is done by a single qubit. A 300 qubit quantum computer can process 2^{300} computations in an instant (more than the number of atoms in the known universe).

Quantum channel

Researchers are often confused over the paradox of dual transmission – encoding via qubits and transmitting encoded information with traditional bits over classical networking. Our communication channels follow laws and constraints of classical physics. Quantum mechanics out-performs these native laws and leads to the emergence of quantum channels. A property related to quantum mechanics, called ‘no-cloning’, states that we cannot clone (construct) an identical copy of the state (spin, orientation or polarisation) of an unknown qubit. The no-cloning property can be considered as a lemma to the ‘Uncertainty Principle’ of quantum mechanics, which formulates the precision inequality in signal processing. Thus, for a hypothetical situation of Alice and Bob, Bob cannot fully decode what Alice had encoded using qubits and dispatched using traditional bits via a classical

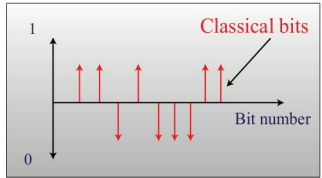


Figure 1: Classical bit

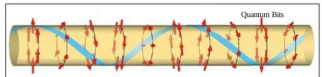


Figure 2: Quantum bit

channel. Even if Bob is aware of the qubit state (a less secure version, only fit for theoretical understanding), the classical channel needs to carry an infinite number of traditional bits so as to fully decode the information encrypted via qubits (as discussed above, a single qubit can have two states, resulting in 2^2 possibilities). Due to these limitations, our current networking media must be replaced by quantum networking to facilitate quantum mechanics in our day-to-day life.

Quantum cryptography

Quantum tips and techniques provide far more secure methodologies for various cryptographic tasks facilitating quantum information theory. Under current commercial communication systems, it is primarily government bodies and a few high-end security companies that are interested in quantum cryptographic techniques. A few firms have started providing networking solutions formulated on quantum mechanics—for example, Swiss Quantum, MagiQ, etc.

One of the most fundamental aspects of any cryptographic system is the key distribution between sender and receiver. Quantum Key Distribution (QKD) is one of the most well established aspects of cryptography, governed by the laws of physics. In our quantum channel, the sender is Alice and the receiver is Bob. Now, as we have seen earlier, qubits follow the ‘no-cloning’ principle. So, if any eavesdropper tries to grasp the state of the qubit (i.e., polarisation) during the secret key exchange between Alice and Bob, the victim qubit is destroyed (either its polarisation, spin or both) and, hence, the eavesdropper is unable to decode the qubit. Bob gets a qubit with a failed checksum (a protocol to check data integrity) and asks Alice for retransmission. Thus, the only overhead is the retransmission of the victim qubit by Alice.

In 1984, Charles Bennett and Gilles Brassard (BB84) formulated a protocol for quantum key distribution. It assumes a quantum channel for key distribution and classical channel